Proof by Induction-Matrices

Key Points

Remember to prove a statement holds for all positive integers by induction, we do the following:

- Show it holds for the base case 1
- Assume it holds for a case k
- Prove the statement then holds for the case $k+1$

For a 2×2 matrix A, an identity of the form of $A^n = \begin{pmatrix} f_1(n) & f_2(n) \\ f_3(n) & f_4(n) \end{pmatrix}$ can be proven by induction. For the inductive step, substitute the assumption into $A^{n+1} = A^n \cdot A$

Basic Skills

Q1. Given the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, calculate

a) A^2 **b)** A^3 **c)** A^4

Q2. Given the matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, calculate

a) A^2 **b)** A^3 **c)** A^4

Q3. Given a matrix A such that $A^n = \begin{pmatrix} 2n+1 & 3n \\ n+5 & 2n-1 \end{pmatrix}$, find

a) A^2 **b)** A^3 **c)** A^{10}

Q4. Calculate $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 4n+2 & n-1 \\ 5 & 2n \end{pmatrix}$

Competency Skills

Q5. Consider the matrix $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. We want to show that $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ using a proof by induction

a) Verify that the formula holds in the case $\mathrm{n} = 1$

b) Assuming that $A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$, show that $A^{k+1} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$

c) Write out the statement we can conclude from parts a) and b) using proof by induction

Q6. Consider the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$. Show that $A^n = \begin{pmatrix} 2^n & 0 \\ 0 & \frac{1}{2^n} \end{pmatrix}$ using a proof by induction

Q7. Consider the matrix $A = \begin{pmatrix} 3 & 1 \\ -4 & -1 \end{pmatrix}$. Show that $A^n = \begin{pmatrix} 1+2n & n \\ -4n & 1-2n \end{pmatrix}$ using a proof by induction

Advanced Skills

Q8. Consider the matrix $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, where a is a constant. Find and prove an expression for A^n

Q9. Consider the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. Show that $A^n = \begin{pmatrix} 1 & n & \frac{1}{2}n(n-1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$ using a proof by induction

Q10. Consider the matrix $A = \begin{pmatrix} 4 & -3 & -3 \\ 3 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}$

a) Diagonalize A by finding a diagonal matrix D and invertible matrix S such that $A = SDS^{-1}$

b) Calculate D^n using a proof by induction

c) Use a) and b) to calculate A^n

Extension

Q11. Consider two sequences $(a_n)_{n=0}^{\infty}, (b_n)_{n=0}^{\infty}$ defined recursively by the formulas $a_{n+1} = 5a_n - 3b_n$, and $b_{n+1} = -6a_n + 2b_n$, with initial terms $a_0 = 1, b_0 = 2$. Find expressions for a_n and b_n

Hint: Can you write the problem in terms of matrices? Also remember the method of Q10.